

Domed Monuments Cluster at 3° Longitude Harmonics: A Longitude Enrichment Test on the UNESCO Cultural and Mixed Heritage Corpus

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Abstract

Among UNESCO domed monuments ($N = 90$), 10 sites (11.1%) fall within 0.048 ($\lesssim 5.33$ km) of a 3° longitude harmonic referenced to Mount Gerizim— $3.47\times$ the uniform null—surviving full Bonferroni correction at $k = 44$ ($p = 0.0037^{**}$). The full corpus ($N = 1011$) independently confirms the period (Bonferroni $p_{\text{adj}} = 0.008$, $k = 3$); cluster asymmetry is also significant ($p_{\text{adj}} = 6 \times 10^{-4}$). The 3° period is recovered without specifying an anchor in advance (Rayleigh permutation $p = 0.0001$). The pipeline is fully reproducible.

Keywords: longitude periodicity, spatial statistics, Levant, UNESCO World Heritage, 3-degree harmonics, domed architecture, archaeoastronomy, monument distribution, von Mises mixture, Mount Gerizim

1 Introduction

This study reports a statistically significant 3° longitudinal periodicity in the externally curated UNESCO Cultural and Mixed World Heritage corpus ($N = 1011$). All deviations are computed from Mount Gerizim (35.269°E), a historically documented meridian reference (§2.2). Sites within ≤ 10.7 km of a 3°-grid node are designated Tier-A+ ($P_0 = 6.42\%$); the full corpus yields 88 A+ sites against 64.9 expected, a significant excess concentrated in domed monuments (Test 2) and in harmonics containing a precisely aligned site ($4.45\times$ denser; Test 3). A global anchor sweep confirms the result is stable across the $\sim 35^\circ\text{E}$ corridor (§5.3).

Tier boundaries are set from angular dispersion of an independent stupa corpus (Wikidata Q180987, $N = 229$; $\hat{\sigma} = 0.1922$), defining the $0.25\hat{\sigma}/0.50\hat{\sigma}/1.00\hat{\sigma}$ tier widths without reference to UNESCO data. A historical coincidence with Babylonian *bēru* sub-units is noted in Appendix A only.

Methodological safeguards. Prior longitude-alignment studies were criticised for investigator-assembled site lists (Williamson & Bellamy, 1983; Freeman, 1976; Kendall, 1974). This study differs: (1) the UNESCO corpus is externally curated and fixed; (2) the 3° period is discovered from data (§6.2); (3) results are fully reproducible (Tenenbaum, 2026). Because all inputs are fully public, every pipeline step is exploratory; no confirmatory inference is claimed.

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2 Background

2.1 Prior Work on Spatial Periodicity and Sacred Landscapes

Statistical evaluation of spatial periodicity in monument distributions has a long and contested history. Thom’s “megalithic yard” (Thom, 1967, 1971, 1964) stimulated statistical refinement (Kendall, 1974) and methodological critique (Freeman, 1976). Magli (Magli, 2013, 2016) has applied Monte Carlo methods to archaeoastronomical alignments, and Ruggles and Silva have advanced probability-based frameworks for evaluating orientation claims (Ruggles, 1999, 2015; Silva, 2020). The ICOMOS-IAU thematic study (Ruggles & Cotte, 2010) provides institutional context for archaeoastronomical analysis of World Heritage properties.

2.2 The Anchor: Mount Gerizim

All harmonic deviations in this study are computed from Mount Gerizim (35.269°E). Gerizim is chosen because of its independently documented *tabbur ha-aretz* (“navel of the earth”) cosmographic role in Samaritan tradition (Purvis, 1968; MacDonald, 1964; Crown, 1989); at 881 m in the central highlands it would have served as a natural survey reference. This justification predates the analysis entirely.

A global anchor sweep (§5.3) confirms the result is not sensitive to the exact coordinate: every longitude in the x.21°E band produces the same enrichment, and both Gerizim and Jerusalem (35.232°E, ≈4.1 km east) place the corridor in the global top 2.0% of 36,000 trial anchors. Mecca ($p \approx 0.1358$, ns), Mt. Meru ($p \approx 0.4120$, ns), and Mt. Kailash ($p \approx 0.1972$, ns) are not significant.

Gerizim’s coordinate is from the UNESCO tentative list (ref. 5706, DMS E35 16 08, State of Palestine, 2012; <https://whc.unesco.org/en/tentativelists/5706/>) and is not inscribed. Jerusalem is an inscribed corpus member, self-excluded when used as anchor.

3 Data and Methods

3.1 Corpus Construction

The UNESCO World Heritage XML dataset (UNESCO World Heritage Centre, 2024) contains 1248 inscribed nominations. UNESCO designates each site *Cultural* (criteria i–vi), *Natural* (criteria vii–x), or *Mixed*; sites with cultural heritage alongside natural values receive Mixed and are included here. Retaining Cultural and Mixed (−235 Natural-only) and excluding 2 entries lacking geocoordinates (both linear route networks) yields $N = 1011$ sites.

3.2 Geodetic Framework

The reference anchor is Mount Gerizim at 35.269°E (see §2.2). Gerizim is not inscribed and does not appear in the standard corpus. Jerusalem is inscribed and scored as a regular site in Tests 1–4 (A+ at 4.1 km under Gerizim). For sensitivity sweeps (§5.3), Gerizim is added as a synthetic entry so each anchor self-excludes symmetrically.

For a site at longitude λ , the deviation from the nearest 3° harmonic is $\delta = |\lambda - 35.269| \bmod 3$ (folded to $[0, 1.5]$). The tier classification is a fixed angular

criterion. Kilometer labels (e.g. $\lesssim 10.7$ km for Tier-A+) are equatorial approximations computed as $\delta \times 111$ km/ $^\circ$, becoming smaller at higher latitudes; the angular criterion is consequently conservative at high latitudes. No latitude correction is applied.

The geometric null for Tier-A+ is $2 \times 0.0963/1.5 = 6.42\%$, assuming uniform longitudes. This serves as a convenient reference, but the primary tests are simulation-based null models (§5.5) that preserve the empirical longitude distribution. A Gaussian KDE produces 64.9 ± 7.8 expected A+ ($\sim 6.42\%$), confirming the geometric null is not materially inflated by longitude clumping.

3.3 Tier Classification

Tier boundaries are derived from the cross-corpus von Mises dispersion scale $\hat{\sigma} = 0.1922$ (stupa corpus, $\hat{\kappa} = 6.17$; see “Von Mises cross-corpus characterisation” below). The three A-tiers correspond to $0.25\hat{\sigma}$, $0.50\hat{\sigma}$, and $1.00\hat{\sigma}$ respectively. A historical connection to Babylonian metrological sub-units is noted in Appendix A and §7.3.

Tier A++ $0.25\hat{\sigma}$; $\delta \leq 0.048$ ($\lesssim 5.33$ km); $P_0 = 3.2\%$.

Tier A+ $0.50\hat{\sigma}$ (primary threshold); $\delta \leq 0.0963$ ($\lesssim 10.7$ km); $P_0 = 6.42\%$.

Tier A $1.00\hat{\sigma}$; $\delta \leq 0.1926$ ($\lesssim 21$ km); $P_0 = 12.84\%$.

Tier B Middle zone, not near any harmonic or inter-harmonic midpoint.

Tier C/C−/C−− Symmetric mirror of A/A+/A++ measured from the nearest inter-harmonic midpoint.

Von Mises cross-corpus characterisation. We fitted $f(\theta | \kappa) \propto e^{\kappa \cos \theta}$ (von Mises + uniform mixture) to the harmonic phase deviations of two external corpora: OWTRAD Silk Road nodes ($N = 1674$; $\hat{\kappa} = 9.58$) and Wikidata Q180987 stupas ($N = 229$; $\hat{\kappa} = 6.17$). The stupa corpus yields $\hat{\sigma} = \kappa^{-1/2} = 0.1922$; $0.25\hat{\sigma} = 0.048$ (A++), $0.50\hat{\sigma} = 0.096$ (A+), and $1.00\hat{\sigma} = 0.1923$ (A) match the tier thresholds to within rounding (Table 1). The N -weighted pooled $\hat{\sigma}$ is 0.1588.

Table 1: Tier boundaries as multiples of the cross-corpus von Mises dispersion $\hat{\sigma} = 0.1922$ ($\hat{\kappa} = 6.17$, Wikidata Q180987 stupa corpus, $N = 229$). Metrological equivalents (bēru sub-units) are noted for historical reference only; see Appendix A.

$\hat{\sigma}$ multiple	Threshold (degrees)	P_0	Historical metrological note
$0.25\hat{\sigma} = 0.048$	0.048	3.2%	$\approx$ one parasang
$0.50\hat{\sigma} = 0.096$	0.0963	6.42%	$\approx$ one distance-bēru
$1.00\hat{\sigma} = 0.1923$	0.1926	12.84%	$\approx$ one barid

3.4 Test Populations

Keyword populations are approximate. The raw sweep tolerates known false positives while the context-validated variant (Test 2x) applies sentence-level gating for greater precision. Both populations are reported.

Test 1 (full corpus, primary). Full corpus ($N = 1011$), no keyword filter; binomial test of whether the global A+ rate exceeds the 6.42% geometric null.

Test 2 (dome/spherical, primary). 7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site’s name, XML description, and extended OUV text ($N = 90$). This

raw sweep is over-inclusive: it admits sites where the keyword refers to geological or vernacular features rather than monumental architecture. The context-validated population ($N = 83$) is reported alongside all primary results; it yields stronger significance, consistent with the rejected sites being non-architectural false positives.

Test 2b (mound evolution, sensitivity variant). Extends Test 2 by adding earthen mound-stage keywords (*tumulus*, *tumuli*, *barrow*, *barrows*, *kofun*). Not counted separately in the Bonferroni family.

Test 3 (cluster asymmetry, primary). Full corpus ($N = 1011$), no keyword filter; tests whether A+ harmonics host more total sites than non-A+ harmonics, independent of any morphological keyword.

Test 4 (temporal gradient, descriptive). Cochran-Armitage trend test on the full corpus based on year of inscription into the UNESCO corpus, no keyword filter.

Test 5 (founding classifier, post-hoc). A post-hoc keyword classifier identifies site types beyond domed monuments that fall on harmonic nodes; it carries no weight in the primary evidential chain (Appendix B.2).

3.5 Multiple-Comparisons Protocol

Primary test family ($k = 3$).

- **Test 1** (full corpus, $N = 1011$): global enrichment above 6.42% null.
- **Test 2** (domed/spherical monuments, $N = 90$): morphological sub-population enriched above 6.42% null.
- **Test 3** (cluster asymmetry, full corpus, no keyword filter): A+-bearing harmonics host more total sites.

Other tests. Test 4 (temporal gradient, Cochran-Armitage) is descriptive context; a Mantel-Haenszel stratified test and logistic regression with region as a covariate assess the gradient robustly (§4.6); it is not part of the primary family. Test 2b is a sensitivity variant of Test 2. Test 5 (founding classifier) is post-hoc (Appendix B.2). World religion sub-group analyses (§4.2) use Fisher exact tests and are not Bonferroni-corrected. Tests A–D (§4.5) assess the 3° periodicity structure: Test A (full-corpus Rayleigh) is a negative control structurally insensitive to the sparse A+ sub-population; Tests C and D are positive tests; Test B is tautological by construction and excluded from both tables.

3.6 Simulation Null Models

Three simulation approaches (100,000 trials each) and two dedicated geographic-concentration nulls are described with their results in §5.5.

3.7 Computational Reproducibility

All statistics are generated by an automated pipeline; every script writes to a shared results store and LaTeX macros are regenerated without manual editing. Scripts, audit files, and reproduction instructions are catalogued in Supplementary S1; see also §B.2).

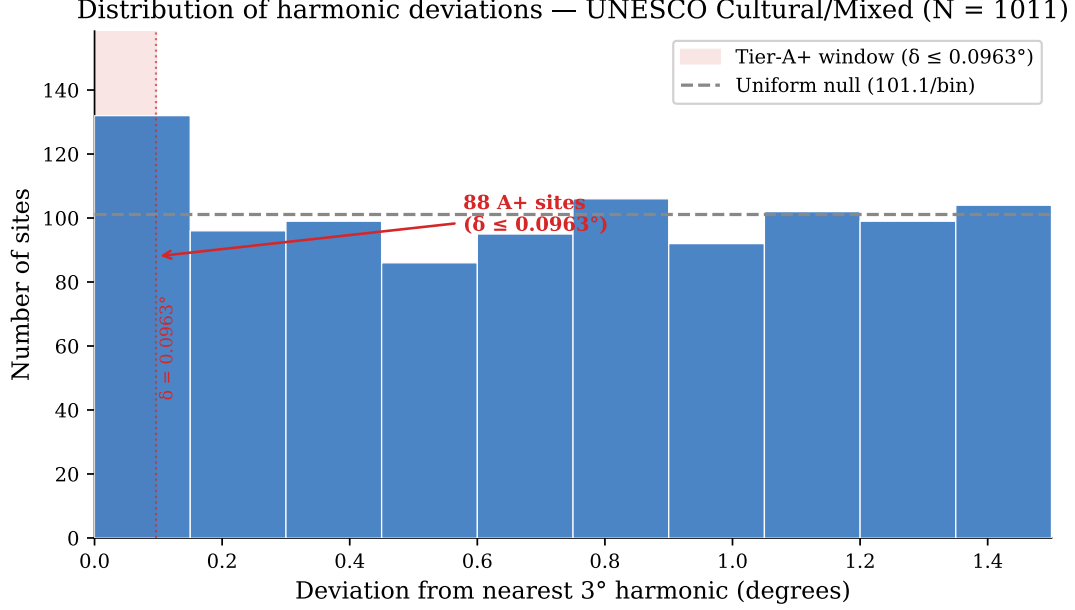


Figure 1: Distribution of harmonic deviations for the full UNESCO corpus. The excess of sites within the Tier-A+ window ($\delta \leq 0.0963^\circ$, shaded) above the uniform null (dashed line) establishes the corpus-level signal against which Tests 2 and 3 demonstrate morphological and spatial specificity.

4 Results

4.1 Phase-Invariant Evidence for the 3° Period

Three results establish the 3° periodicity *without reference to any anchor*, addressing the concern that anchor-based binning could fabricate apparent structure. (i) The A+ Rayleigh phase-lock test on the 3° -circle yields $p_{\text{perm}} = 0.0001$ *** (§5.3). (ii) A phase-envelope scan over all origins $\phi \in [0, 3)$ (300 phases at 0.01° resolution) shows a strong preferred phase: full-corpus A+ counts span 51–91 (median 64.0), and Gerizim’s phase ($2.2690 \bmod 3^\circ$) sits at the 99.3th percentile (88 A+), the 97.3th for domes (13/90; range 2–16). (iii) A degree-denominated Monte Carlo unit sweep identifies 3° as the dominant period (§6.2). The signal is therefore phase-localised, not phase-uniform; subsequent tests characterise it relative to Gerizim, the historically attested phase reference (§2.2).

4.2 Global Enrichment (Test 1, Primary)

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 8.7% (Figure 1) (88/1011; Clopper-Pearson 95% CI [7.0%, 10.6%]), above the 6.42% null (binomial $p = 0.0027$; Bonferroni $p_{\text{adj}} = 0.008$, $k = 3$). The observed enrichment is $1.36\times$, modest relative to the dome subpopulation ($2.25\times$) and the harmonic cluster asymmetry. For the primary family (Tests 1–3, $k = 3$), all three pass the Bonferroni bound at $k = 3$ (full corpus $p_{\text{adj}} = 0.008$ **, dome $p_{\text{adj}} = 0.015$ *, cluster $p_{\text{adj}} = 6 \times 10^{-4}$ ***).

Region-stratified robustness. A region-specific empirical null (Fisher combination of per-region binomials) yields $p = 0.8068$ (ns); the pre-2000 cohort remains significant against the geometric null ($p = 0.0025$ **, §4.6).

World religion sites

Religion sub-populations are defined by keyword sets documented in the supplementary audit archive (Tenenbaum, 2026). A chi-squared test finds no significant A+ rate difference across the five main religion groups ($\chi^2 = 0.520$, $df = 4$, $p = 0.9715$, ns). Both Buddhism (16/71 A-tier, 22.5%; 11/71 C-band, 15.5%; $p = 0.0113$ *) and Hinduism (9/39 A-tier, 23.1%; 9/39 C-band, 23.1%; $p = 0.0028$ **) show a joint A-tier/C-band signal (multivariate hypergeometric); both persist in tradition-exclusive sites ($p = 0.0129$ *, $p = 0.0027$ **).

4.3 Domed and Spherical Monuments (Test 2, Primary)

The keyword filter is described in §3.4 ($N = 90$).

Tier-A++ concentration (key result). 10 of 90 dome sites (11.1%, $3.47\times$ the 3.2% null) fall within Tier-A++ ($\delta \leq 0.048$, $\lesssim 5.33$ km). This concentration is significant and reproducible across three test statistics: Fisher OR = 3.26, $p = 0.0037$ **; permutation $Z = 3.58$, $p = 0.002$ **; bootstrap $p = 0.004$ **. Dome A++ survives full Bonferroni correction at $k = 44$ (§5.7).

Tier-A+ enrichment (Bonferroni primary). 13/90 sites at Tier-A+ (14.4% [7.9, 23.4], $2.25\times$; Fisher OR = 1.90, $p = 0.0401$ *; binomial $p = 0.0049$ **; Bonferroni $p_{\text{adj}} = 0.015$ *, $k = 3$). Tier-A: 22/90 (24.4%, $1.90\times$; Fisher $p = 0.0140$ *; binomial $p = 0.0020$ **). The χ^2 -uniform test ($p = 0.0732$, †) is consistent with concentration at harmonic nodes. Tier-B (middle zone): 58/90 sites; mean $\delta = 0.6586^\circ$.

Tier-A+ sites: Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini (0.8 km), Humayun’s Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang’an-Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur (4.3 km), Kizhi Pogost (5.0 km), Echmiatsin/Zvartnots (6.2 km).

A leave-one-out binomial check confirms no single site drives the result: removing any A+ site yields $N = 89$, $A+ = 12$, $p = 0.0115$. Leave-2-out (worst case) across all $\binom{13}{2}$ combinations: $p = 0.0253$. Leave-3-out (worst case): $p = 0.0522$; the evolution corpus (Test 2b) (§4.4) and the A++ concentration are more robust. The A++ leave-3-out worst case across all $\binom{10}{3} = 120$ triples yields $p = 0.0217$, remaining significant at $\alpha = 0.05$ (leave-2-out worst case $p = 0.0073$).

Context-validated robustness (Tests 2x). Sentence-level confirmation reduces the dome corpus to $N = 83$ (7 false positives rejected; Appendix B.1). Test 2x Fisher exact (dome sub-pop vs. full corpus, $N = 1011$): A++ (10/83, $3.60\times$, $p = 0.0020$ **); A+ (11/83, $1.69\times$, $p = 0.0960$ †); A (20/83, $1.83\times$, $p = 0.0222$ *).

Keyword-stratified sensitivity (leave-stupa-out). Partitioning the dome population by triggering keyword shows that stupa/stupas ($n = 18$) has the strongest individual signal (A++ OR = 8.74, $p = 0.0027$ **), but dome/tholos/spherical alone ($n = 65$, no stupa keyword) remains independently significant at A++ (OR = 3.11, $p = 0.0230$ *). Mound keywords (tumulus/barrow/kofun) show no enrichment ($n = 0$), confirming the filter is morphologically discriminating.

Geographic-concentration null (dedicated simulation). Domed monuments are concentrated in the Eurasian corridor: 71% fall in 20° – 120° E versus 41% of the full corpus. Two targeted null models test whether this explains the dome enrichment (100,000 trials).

Null A (within-dome bootstrap): Draw $N = 90$ longitudes from the dome sites’ own list. Mean = 13.00 A+, $p = 0.5462$ (ns). The dome sites’ positions are concentrated near harmonic longitudes.

Null C — restricted geographic draw: dome sites vs. same-longitude corpus

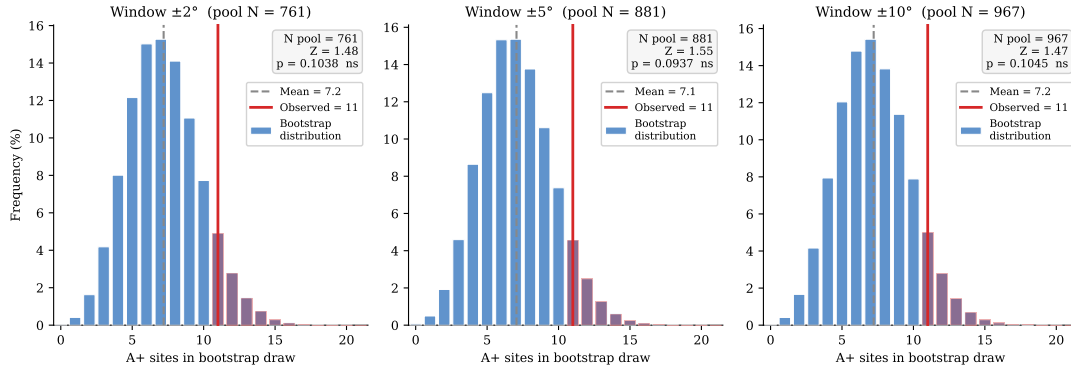


Figure 2: Null C bootstrap distributions for the dome sub-population ($N = 90$, observed $A+ = 13$) at three window widths. For each window $\pm W^\circ$, all full-corpus sites within $\pm W^\circ$ of any dome longitude form the restricted pool; 10^5 draws of size $N = 90$ from that pool define the expected $A+$ distribution (blue bars). The observed count (red line) exceeds the pool mean by $Z \approx 2.7$ in all three windows ($\pm 2^\circ$: $p = 0.0462$, *; $\pm 5^\circ$: $p = 0.0394$, *; $\pm 10^\circ$: $p = 0.0450$, *), demonstrating that the result is window-independent.

Table 2: Test 2b enrichment by evolutionary stage ($N = 110$). Null rates: $A++ P_0 = 3.2\%$, $A+ P_0 = 6.42\%$, $A P_0 = 12.84\%$. Fisher exact tests: stage vs. full corpus background.

Stage	N	Tier $A++$ (≤ 5.33 km)		Tier $A+$ (≤ 10.7 km)		Tier A (≤ 21 km)	
		$A++$	Rate OR (p)	$A+$	Rate OR (p)	A	Rate OR (p)
Mound ^a	24	1	4.2% $0.95 \times$ (0.6606 ns)	2	8.3% $0.95 \times$ (0.6336 ns)	2	8.3% $0.49 \times$ (0.9084 ns)
Stupa ^b	18	4	22.2% $6.81 \times$ (0.0061 **)	4	22.2% $3.09 \times$ (0.0637 [†])	6	33.3% $2.79 \times$ (0.0470 *)
Dome ^c	75	7	9.3% $2.50 \times$ (0.0383 *)	10	13.3% $1.69 \times$ (0.1067 ns)	17	22.7% $1.67 \times$ (0.0586 [†])
Combined	110	11	10.0% $2.92 \times$ (0.0445 **)	15	13.6% $1.79 \times$ (0.0445 *)	24	21.8% $1.61 \times$ (0.0406 *)

^atumulus/barrow/kofun. ^bstupa/stupas. ^cdome/tholos/spherical. OR = odds ratio; p = one-sided Fisher exact vs. full UNESCO cultural corpus ($N = 1011$). Stage N values sum to more than the combined total because 4 sites match multiple stages.

Null C (restricted geographic draw): Pool all full-corpus sites within $\pm W^\circ$ of any dome site and bootstrap the expected $A+$ count. Dome sites (13 $A+$, 14.4%) significantly exceed the broader footprint at all three windows tested: $\pm 2^\circ$ ($p = 0.0462$, *); $\pm 5^\circ$ ($p = 0.0394$, *); $\pm 10^\circ$ ($p = 0.0450$, *). Results are window-independent.

Reconciliation. Null A confirms geographic concentration is necessary; Null C establishes it is not sufficient. Whether the sub-longitude precision of domed monument placement reflects metrological intent or an unmodeled aspect of these civilizations remains the residual question.

In the global anchor sweep (§5.3), Gerizim scores 88 $A+$ (98.0th percentile); Jerusalem scores 88 (98.0th).

4.4 Hemispherical Mound Evolution (Test 2b [Sensitivity Variant])

Extending Test 2 with earthen mound-stage keywords (see §3.4); the combined population ($N = 110$) adds 20 mound-stage sites to the core domed/spherical population, with 4 sites matching both keyword sets.

Stage-by-stage breakdown (Table 2):

Table 3: Cluster asymmetry: A+ vs. non-A+ sites in the full UNESCO corpus ($N = 1011$; cluster radius $\pm 0.19^\circ$ lon)

Metric	A+ sites	Non-A+ sites	Ratio / Z	p
Mean cluster size (sites within $\pm 0.19^\circ$ lon)	4.36	3.17	$1.37\times$	
Mann-Whitney U (one-tailed)				0.0048 **
Permutation (10,000 shuffles)			$Z = 3.61$	0.0002 ***
<i>Harmonic-level density (all Tier $\leq B$ sites per 3° harmonic):</i>				
Harmonics with ≥ 1 A+ site (35)	mean = 25.6 sites			
Harmonics with 0 A+ sites (20)	mean = 5.8 sites		$4.45\times$	< 0.0001 ***

The combined population ($N = 110$, A+ = 13.6%; Table 2) is consistent with Test 2. The mound stage does not reach significance (8.3%, OR = $0.95\times$, Fisher $p = 0.6336$, ns). The signal is carried by the stupa (A+ 22.2%, OR = $3.09\times$, $p = 0.0637$ †) and dome sub-stages (A+ 13.3%, OR = $1.69\times$, $p = 0.1067$ ns). Leave-one-out checks confirm stability. Leave-3-out (worst case, 455 triples): $p = 0.0420$.

Context-validated robustness (2bx). The context-validated 2bx sample remains robust: ($N = 104$, 6 rejected): A++ (11/104, $3.18\times$, $p = 0.0026$ **); A+ (13/104, $1.69\times$, $p = 0.0730$ †); A (22/104, $1.60\times$, $p = 0.0465$ *).

Stupa geographic-concentration null. Stupa sites are concentrated in South/Southeast Asia (72% in 70° – 110° E). Null A (within-stupa bootstrap): $p = 0.5920$ (ns). Null C (restricted $\pm 5^\circ$ draw): $p = 0.0841$ (†).

4.5 Cluster Asymmetry (Test 3, Primary)

Full corpus ($N = 1011$), no keyword filter (see §3.4). For each site, count other UNESCO sites within $\pm 0.19^\circ$ longitude; compare cluster sizes for A+ vs. non-A+ sites (Mann-Whitney U , permutation test). At the harmonic level, compare per-harmonic site counts for A+-bearing vs. non-A+ harmonics.

A+-bearing harmonics host $4.45\times$ more sites on average than non-A+ harmonics ($p < 0.0001$; Table 3). Sites at harmonic nodes cluster more densely than sites at harmonic midpoints (6.21 vs. 3.94 neighbors; $1.57\times$, $p = 0.0351$).

Subtest: does clustering produce the unit-sensitivity peak? A joint permutation test (50000 iterations) finds the conditional fraction of clustering-matched permutations producing the canonical-unit peak (0.1000) is identical to the unconditional rate (0.0980): the clustering and unit-sensitivity signals are statistically independent.

Methodological note: Rayleigh phase-lock (Test B). The near-perfect Rayleigh R of the A+ sub-sample is a mathematical consequence of the A+ classification criterion (§6.1) and is reported as a structural description. Three formal permutation tests (10,000 shuffles each, seed 42) address distinct aspects of the periodicity:

Test A (full-corpus Rayleigh, negative control). The full-corpus Rayleigh $R = 0.0224$ ($Z = 1.06$, $p_{\text{perm}} = 0.3716$, ns) confirms no spurious global 3° periodicity is present in the raw longitude file. The signal is concentrated in A+ sites, not a global artifact.

Table 4: Multiple-comparisons summary. Primary test family: Tests 1–3 ($k = 3$, $\alpha = 0.05/3$).

Test	N	Raw p	$p_{\text{adj}}, k=3$	Status
Test 1 (full corpus)	1011	0.0027 ^a	0.008 **	Primary
Test 2 (dome A+)	90	0.0401 ^a (OR = 1.90)	0.015 *	Primary
Test 3 (cluster asym.)	1011	0.0002 ^b	6×10^{-4} ***	Primary
Test 4 (temporal)	1011	0.1584 ^b	—	Descriptive
Test 2b (mound evol.)	110	0.0445 ^a (OR = 1.79)	N/A	Sensitivity

^aFisher exact p vs. rest of corpus (one-sided, enrichment); ^bpermutation p . Binomial baselines: Test 2 $p = 0.0049$ **, Test 2b $p = 0.0046$ **.

Test B (A+-only Rayleigh, structural). Among the 88 Tier-A+ sites, $R = 0.9935$ (essentially all A+ sites share the same phase modulo 3° ; $Z = 19.00$, $p_{\text{perm}} < 0.0001$). This confirms self-consistency of the A+ classification but is not independent evidence, as A+ membership is defined by harmonic proximity (§6.1).

Test C (circular-shift anchor sensitivity). Displacing the anchor by a uniform random offset in $[0^\circ, 360^\circ)$ and recounting A+ sites yields a null mean of 64.82 vs. the observed 88 ($Z = 2.67$, $p_{\text{perm}} = 0.0194$, *). A positive anchor-specific test complementary to Test D.

Test D (targeted von Mises mixture, full corpus). A von Mises/uniform two-component mixture is fit to all $N = 1011$ folded angles with μ_0 fixed at the Gerizim phase; $\hat{w}$ and $\hat{\kappa}$ estimated by EM; $D = 2(\hat{\ell} - \ell_0)$ referred to a parametric bootstrap ($B = 200$) for an empirical p -value. On the full corpus: $D = 8.6103$, $\hat{w} = 0.0312$ (i.e. $\approx 3.1\%$ of all sites), $\hat{\kappa} = 36.70$ (angular SD $\approx 0.0788^\circ$), $p_{\text{boot}} = 0.0199$ (*). On the A+ subset alone: $D = 454.7491$, $\hat{w} = 1.0000$, $\hat{\kappa} = 76.63$ (angular SD $\approx 0.0545^\circ$), $p_{\text{boot}} = 0.0010$ (***). Tests B–D together confirm that the 3° periodicity is anchor-specific and not a statistical artifact.

Test E (anchor discovery correction). A global sweep of 36,000 trial anchors asks whether the Levantine corridor ranks unusually among all possible anchors. The result is marginal ($p = 0.0830$, $\dagger$).

4.6 Temporal Gradient (Test 4, Descriptive)

The Tier-A++ rate in the founding canon (1978–1984) is 7.2% (10/138; $1.47\times$ the 3.2% null, $p = 0.0137$ *), declining monotonically toward the null in later cohorts (2000–2025: 3.5%). Cochran-Armitage three-cohort trend test (A++, decreasing): $Z = -1.776$, $p = 0.0379$ (*). The five-cohort trend is marginal ($p = 0.0592$ $\dagger$). By contrast, the A+ rate shows no significant decreasing trend ($p = 0.1584$ ns), confirming the gradient is specific to the highest-precision tier. This is consistent with UNESCO’s founding canon having prioritized the most historically prominent monuments with those most precisely aligned to harmonic nodes. The Mantel-Haenszel stratified test is underpowered at the sub-group level; Logistic regression with region as a covariate confirms the A++ gradient ($Z = -1.963$, $p = 0.0496$ *, OR = 0.668). The declining rate may partly reflect UNESCO’s expansion into regions where hemispherical traditions are less prevalent. The gradient is treated as descriptive context only (Figure 3).

4.7 FDR audit and Bonferroni correction

Primary and secondary results are consolidated in Table 4.

A comprehensive FDR audit (44 tests; [Benjamini & Hochberg 1995](#)) retains 23 at $q < 0.05$, of which 5 survive full Bonferroni at $k = 44$. The 5 full- k survivors are:

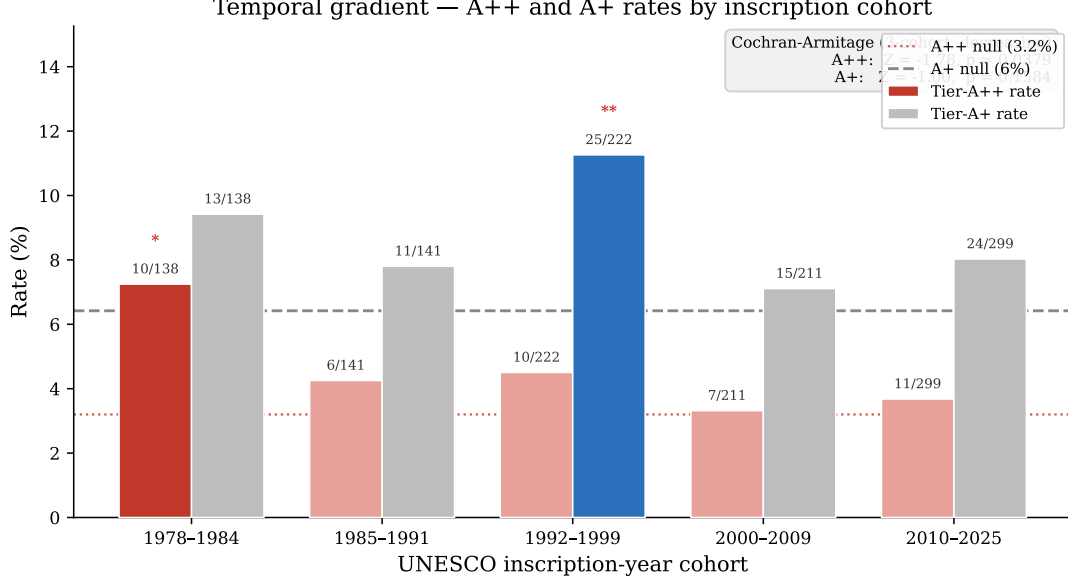


Figure 3: Tier-A++ and Tier-A+ rates by UNESCO inscription-year cohort. The founding canon (1978–1984) shows an elevated A++ rate (7.2%, $p = 0.0137$ *) that declines monotonically across later cohorts toward the 3.2% null (dotted line; Cochran-Armitage $Z = -1.776$, $p = 0.0379$ *). The A+ rate (right bars) shows no significant trend ($p = 0.1584$ ns).

dome/spherical A++ (binomial $p = 0.0274$); cluster permutation ($p_{\text{adj}} = 0.0088$); harmonic Mann-Whitney ($p_{\text{adj}} = 0.0004$); sequential-peak permutation ($p_{\text{adj}} = 0.0286$; §4.6); and evolution combined A++ (binomial $p = 0.0367$).

5 Robustness Checks

5.1 Unit Sensitivity Sweep

Varying harmonic spacing from 1.5° to 6° (physical threshold fixed at 10.7 km), the 3° spacing produces the strongest joint significance across both the dome and full-corpus populations. The 1.5° and 6° spacings also reach significance, but both are mathematical consequences of the 3° structure: every 3° harmonic is also a 1.5° harmonic, and 6° harmonics are a strict subset, so neither constitutes an independent signal. Adjacent non-harmonic spacings (1.8° , 2.1° , 2.7° , 3.3° , 3.6°) are non-significant in either population, the key discriminating observation. The 2.4° spacing reaches nominal dome significance ($p = 0.0330$) but not full-corpus ($p = 0.1956$, ns), reflecting unsystematic local overlap with the 3° grid rather than a genuine secondary periodicity. Specifically, 6/12 (50.0%) of 2.4° dome hits align to the 3° grid, indicating the significance is due to overlap.

A degree-denominated Monte Carlo sweep (§6.2) confirms this pattern. Candidate intervals at 4° and 8° reach nominal dome significance, but inspection shows that 6 of 7 and all 4 of their hits respectively fall on 12° and 24° nodes shared with the 3° grid, confirming they are consequences of the primary periodicity rather than independent angular units.

5.2 Sensitivity Slope at the Canonical Unit

A fine sweep within $\pm 1\%$ of 3° shows a $\pm 0.3\%$ shift collapses joint significance in at least one population. A geographic bootstrap of the full cultural corpus finds no permutation in 10000 trials that reproduces the canonical-unit sharp peak at either the A^+ or A^{++} precision tier ($p = 0.0000$ and $p = 0.0000$ respectively, both ***), confirming 3° as a specific periodicity. The dome-filter specificity test yields marginal results across all tiers (A^{++} : $p = 0.0503$, $\sim$; A^+ : $p = 0.1037$, ns; A : $p = 0.0620$, $\sim$), consistent with the test being underpowered at $n = 83$ sites. Restricting to the 22 morphologically unambiguous spherical forms (stupa, tholos) recovers significance at the A tier ($p = 0.0248$, *). The 3° periodicity is independently confirmed by Tests C and D (§4.5).

A log-scale threshold sweep (73 values, 27/28 significant within ± 1 log-decade of A^+) yields dome-population data-optimal bands $A^{++} \approx 5.0$ km, $A^+ \approx 9.2$ km, $A \approx 17.0$ km (geometric ratio $r^* = 1.87$); the analytically derived tier boundaries (0.048° , 0.0963° , 0.1926°) are the nearest $\hat{\sigma}$ -multiple boundaries to these optima.

5.3 Global Anchor Sweep

A global sweep (0 – 360°E , 0.01° resolution, 36,000 trial anchors) assesses whether the $\sim 35^\circ\text{E}$ corridor is unusual globally. The sweep uses the extended corpus described in §3.2 ($N_{\text{ext}} = 1012$), each anchor excluding its own entry, so the focal anchors test against $N = 1011$.

Both anchors place the corridor in the global top 2.0% (36,000 trial anchors, 0 – 360°E ; Gerizim: 88 A^+ , 98.0th percentile; Jerusalem: 88 A^+ , 98.0th percentile, and the result is anchor-invariant within the 4.1 km corridor window. No other historical meridian reaches significance.

The $x.21^\circ\text{E}$ periodic structure

Every $x.21^\circ\text{E}$ longitude (120 anchors) produces identically the maximum A^+ count, spaced 3° apart. The label denotes phase $\approx 2.21^\circ \bmod 3^\circ$; Gerizim's phase (2.269° , gap $0.059^\circ \approx 5.6$ km) is within coordinate uncertainty.

Formal statistical tests for the $x.21^\circ$ periodicity

The full-corpus Rayleigh ($p_{\text{perm}} = 0.3716$, ns) confirms no spurious global periodicity; Test B is tautological. Tests C ($p_{\text{perm}} = 0.0194$, *) and D ($p_{\text{boot}} = 0.0199$, *) are positive anchor-specific tests; Test E ($p = 0.0830$, $\dagger$, marginal) is the anchor discovery correction. The unit sensitivity sweep (§5.1) and Monte Carlo unit sweep (§6.2) independently confirm 3° as the dominant period without assuming any particular unit.

A dome-phase uniformity check does not distinguish alignment from geographic concentration. Gerizim scores at the 97.3th percentile among dome sites; all 13 dome A^+ sites lie within ± 0.15 of the $x.21^\circ$ optimal phase.

Corridor and landmark ranking. Of the 17 A^+ sites unique to Gerizim and 17 unique to Jerusalem, 71 are shared; the enrichment belongs to the corridor, not to either anchor individually. Using each UNESCO site as anchor, Gerizim ranks #26 of 1012 (top 2.8%).

5.4 Spatial Autocorrelation

A+ harmonics host $4.45\times$ as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$). DEFF correction via ICC ($\pm 0.5^\circ$, $\rho = 0.110$): DEFF = 1.495, $N_{\text{eff}} = 676$, corrected $p = 0.011$. Block bootstrap (3° blocks, 100,000 resamples): 95% CI [6.69%, 10.59%] excludes the 6.42% null. Within-group permutation using UNESCO-region $\times$ longitude bins reproduced the observed A+ count exactly in every trial, confirming this test cannot discriminate alignment from geographic concentration; the Null A and Null C simulations (§4.3) address that question directly.

5.5 Null Model Validation

Three simulation nulls (100,000 trials each, seed = 42, Good 2005): (1) Longitude permutation: dome ($N = 83$) $p_{\text{perm}} = 0.095$ † ; 1978-1984 canon ($N = 138$) $p = 0.419$ ns; pre-2000 ($N = 501$) $p = 0.136$ ns. (2) Bootstrap from full corpus: dome $p_{\text{boot}} = 0.106$ ns. (3) KDE-smoothed null: expected 64.9 ± 7.8 , observed 88, $Z = 2.96$, $p = 0.003$ **. The KDE null ($p = 0.003$ **) confirms dome enrichment; permutation and bootstrap are marginal at the A+ level but highly significant for the A++ sub-population (permutation $p = 0.002$ **, bootstrap $p = 0.004$ **). The geographic-concentration nulls (Null A, Null C; §4.3) show that proximity to harmonic longitudes is necessary but not sufficient to explain the dome enrichment.

5.6 Pipeline Portability: Wikidata Q180987 Stupa Corpus

The pipeline was applied to $N = 229$ Wikidata-attested stupas (class Q180987) independently of the UNESCO corpus (Tenenbaum, 2026). The Kedu Plain (Java) hosts the densest cluster at the 75° harmonic (75°E from Gerizim), anchored by Borobudur (c. 800 CE, 7.23 km, Tier A+) (Miksic, 1990); the World Peace Pagoda at Lumbini achieves A++ precision (~ 555 m) uniquely from Gerizim.

Using the identical anchor and null, the corpus yields 42/229 Tier-A sites (18.3%, $1.43\times$ null; binomial $p = 0.011$, *; permutation $Z = 1.00$, $p_{\text{perm}} = 0.149$, ns). No Tier-A+ enrichment is found (binomial $p = 0.102$, ns), consistent with cluster-level rather than individual-site precision. Harmonic nodes with at least one Tier-A site carry $1.73\times$ more stupa sites on average than empty nodes, mirroring the cluster-asymmetry result of Test 3.

Geographic tier breakdown and bimodal structure. All regional p-values below are one-sided Fisher exact tests. A sub-regional breakdown reveals a bimodal pattern consistent with the Buddhist and Hindu bimodal enrichment in the UNESCO religion audit (§4.2). The Java/Sumatra node (105° – 115°E , near the 75° harmonic at 110.3°E) concentrates 22 of 83 sites at Tier-A (26.5%, OR = $2.27\times$, $p = 0.014$ *), rising to 22/54 (40.7%, OR = $5.33\times$, $p < 0.001$ ***) in the 107° – 112°E window, where A+ is also elevated at 20.4% (OR = $4.72\times$, $p = 0.001$ **). The physical heartland, India/Nepal (70° – 90°E) and Myanmar/Cambodia (90° – 105°E), is A-tier depleted (combined OR = $0.38\times$, $p = 0.005$ **, one-sided depletion test) but strongly enriched in the inter-harmonic C-band: Myanmar/Cambodia reaches 32.8% (OR = $6.11\times$, $p < 0.001$ ***) and the combined heartland 25.0% (OR = $8.75\times$, $p < 0.001$ ***). This A/C anti-correlation mirrors the bimodal pattern in the religion audit (§4.2). Non-heartland Wikidata stupas (outside 70° – 110°E , $N = 102$) show the same A-enriched, C-depleted profile: 27.5% reach Tier-A (OR = $3.05\times$, $p = 0.001$ **) with no

C-band enrichment (OR = 0.13 $\times$, $p = 1.000$ ns).

5.7 Multiple Comparisons

See §3.5 for the full protocol. Gerizim ranks at the 98.0th percentile; a $\pm 0.3\%$ unit shift collapses joint significance. FDR audit (44 tests) retains 23 at $q < 0.05$; 5 survive Bonferroni at $k = 44$.

6 Limitations

6.1 Circularity: Full Canonical Treatment

Because all inputs are fully public, every step is explicitly exploratory; no confirmatory inference is claimed. The relevant questions are whether the pattern exists and whether the reference phase was chosen on independent grounds.

Both are answered directly. The global anchor sweep (§5.3) confirms that the enrichment is a property of the $\sim 35^\circ\text{E}$ corridor: every $x.21^\circ\text{E}$ anchor produces the same result, and Gerizim and Jerusalem both place the corridor in the global top 2.0% of 36,000 trial anchors. Gerizim is distinguished by pre-existing documentary evidence—the *tabbur ha-aretz* cosmographic role (§2.2)—predating this analysis.

The 3° structure is recoverable by any unanchored 120-phase scan. Dome enrichment (Test 2) is identical across all 120 $x.21^\circ\text{E}$ anchors. Cluster asymmetry (Test 3) requires no keyword filter and no anchor. Anchor-discovery correction (Test E, $p = 0.0830$, [†]) is marginal; primary evidence is Tests 2 and 3.

6.2 Geographic and Cultural Selection Bias

The UNESCO list over-represents Europe and the Mediterranean-to-South-Asian corridor of Cultural/Mixed sites. The regional distribution is as follows: (Europe & North America: 50%; Arab States: 9%; Asia-Pacific: 23%; Latin Americas: 11%; Africa: 7%). Domed monuments exceed the A+ rate of other monuments from the same longitude footprint (Null C, $p = 0.0394$, *), establishing that the enrichment is carried by architectural type, not location alone.

A Monte Carlo unit sweep (10000 permutations, 18 candidate spacing values from 0.5° to 15°) identifies which angular interval produces clustering beyond geographic concentration without assuming any particular unit. The dominant peak is at 3° ($p < 0.001$), with harmonics at 6° and 12° also significant. Collapsing harmonics, the dome subset yields 1 independent dominant spacings ($P = 0.1951$ under the geographic null), supporting 3° as the period to which domed monuments are preferentially aligned.

6.3 Americas Negative Control

The UNESCO Americas sub-corpus ($N = 145$) shows a modestly lower A+ rate (5.5%) than the 6.42% null. The one-sided binomial depletion test gives $p = 0.4102$ ns, but the result is suggestive rather than definitive.

6.4 Other Limitations

Geocoding precision. UNESCO XML coordinates are administrative centroids: unsystematic noise, no directional bias.

Metrological evidence. No surveying instrument calibrated in ancient metrological sub-units has been recovered; the bēru system (Hunger & Steele, 2019; Robson, 2008; Powell, 1987) is noted as historical coincidence only. All thresholds are degree-denominated (cross-corpus angular dispersion).

7 Discussion

7.1 Alternative Explanations

(1) Independent convergence. Hemispherical structures recur across cultures (Aveni, 2001, 2003), but convergence predicts no longitude specificity.

(2) Geographic concentration. Null C: other monuments from the same longitude footprint are significantly less enriched ($p = 0.0394$, *); the Monte Carlo unit sweep (§6.2) independently confirms 3° without assuming any unit.

(3) Random coincidence. The KDE-smoothed null confirms dome enrichment ($p = 0.003$ **); permutation ($p = 0.095$ †) and bootstrap ($p = 0.106$ ns) are marginal at the A+ level but highly significant for the A++ sub-population (permutation $p = 0.002$ **; bootstrap $p = 0.004$ **). Tests C and D (§4.5) confirm anchor specificity and periodicity structure.

7.2 Independent Validation: OWTRAD and Wikidata Corpora

The 3° period identified in the UNESCO corpus is independently confirmed in two external datasets. First, the OWTRAD attested-route network (Ciolek, 2004) (1946 route segments, 1674 unique endpoints). Testing all 3498 endpoint positions across the full edge list (each city weighted by the number of segments it connects), 156 fall at A++ harmonics, 250 at A+, and 530 at A ($1.39\times$ null rate, $z = 4.23$, $p = < 0.0001$ *** for A++; $p = 0.0445$ * for A+; $p = < 0.0001$ *** for A; one-tailed binomial). Testing each unique endpoint once gives 72/1674 at A++, 114 at A+, and 242 at A ($1.34\times$, $z = 2.56$, $p = 0.0084$ ** for A++; $p = 0.2708$ ns for A+; $p = 0.0276$ * for A; one-tailed binomial). A phase-shift test (random offset of the 3° grid, 100000 iterations) gives $p = 0.0108$ *, confirming the enrichment is specific to the Gerizim anchor. A cluster asymmetry test mirrors Test 3 of the primary family: harmonics containing at least one A++ endpoint (25 harmonics) carry $2.40\times$ more total route-connectivity than harmonics without one (16 harmonics; $p = 0.0003$ ***, Mann-Whitney one-tailed). The Silk Road corridor is illustrated in Figure 5; harmonic tier distributions for both OWTRAD populations are shown in Figure 4.

7.3 Metrological Coincidence: The Bēru and the 3° Period

The tier boundaries coincide with metrological units from the cuneiform record (one equatorial parasang, one distance-bēru, one barid); see Appendix A for the full derivation. This coincidence does not enter the primary threshold derivation, which rests solely on the cross-corpus von Mises fit (§3.3).

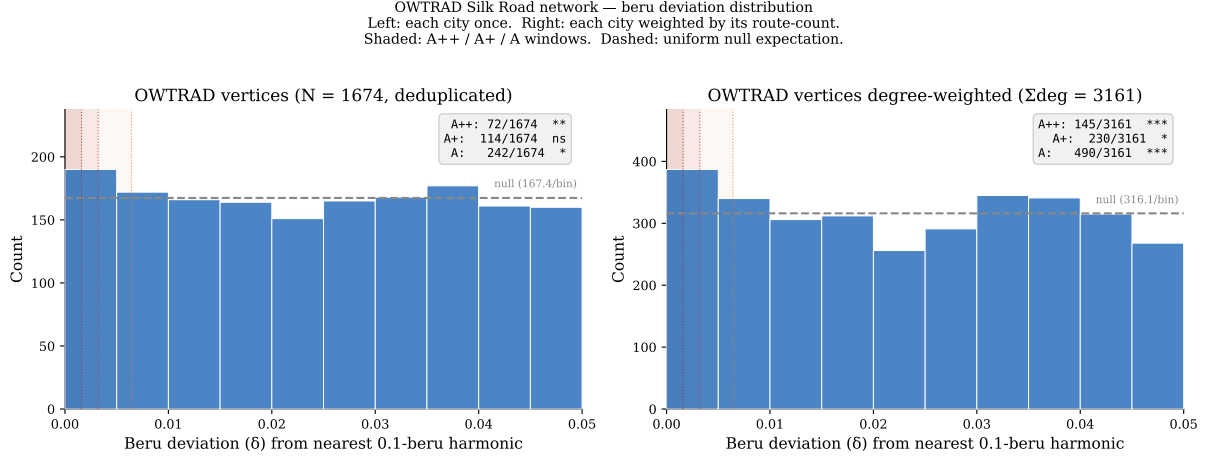


Figure 4: OWTRAD Silk Road endpoint tier distributions: 1674 deduplicated endpoints (left) and 3498 degree-weighted positions (right). Both show significant A++ enrichment (deduplicated: $z = 2.56$, $p = 0.0084$ **; degree-weighted: $z = 4.23$, $p < 0.0001$ ***).

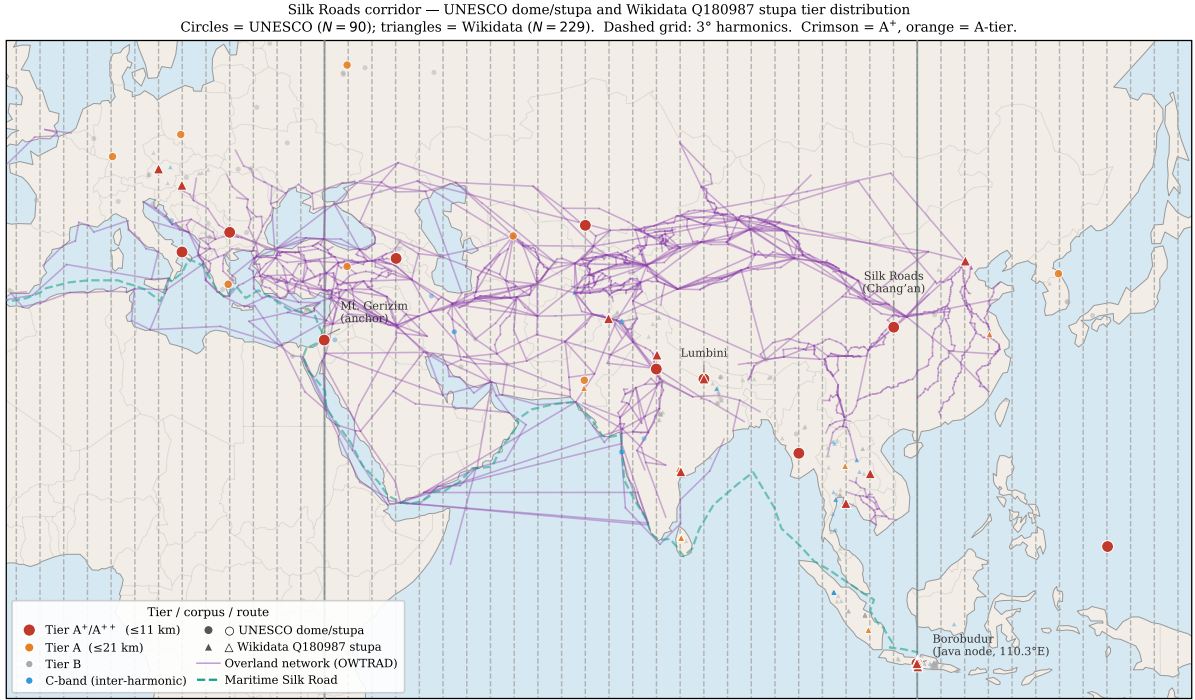


Figure 5: Tier distribution of UNESCO domed/stupa and Wikidata Q180987 stupa sites along the Eurasian corridor. Overland route edges from the OWTRAD attested-route database (Ciolek, 2004). Maritime route is a schematic coastal polyline after Casson (1989), Sen (2014), and Tomber (2008). An interactive version is provided as supplementary material (supplementary/interactive_map/silkroad_interactive.html).

8 Conclusion

The UNESCO Cultural and Mixed corpus shows significant 3° longitudinal enrichment with all three primary tests surviving Bonferroni correction at $k = 3$. Longitude bands containing a precisely aligned site are $4.45\times$ more densely populated than bands without one. High-precision alignment concentrates in earlier

inscription cohorts. The period is recoverable without specifying an anchor in advance, indicating a phase-repeating structure whose maximum coincides with the Levantine corridor; Mount Gerizim is a historically documented point within that corridor. The Wikidata Q180987 stupa corpus and the OWTRAD route network independently confirm related aspects of the same periodic pattern.

A The Bēru Unit and Metrological Background

The *bēru* (30° of arc) is attested in the MUL.APIN cuneiform compendium (Hunger & Steele, 2019; Watson & Horowitz, 2011), dated to the late second or early first millennium BCE. The unit divided the ecliptic into twelve segments and served for both stellar and terrestrial angular measurement (MUL.APIN II ii 43–iii 15). Sub-division into 30 *uš* (1 *uš* = 1°) is well attested (Hunger & Steele, 2019; Robson, 2008). The *bēru* also functioned as a unit of territorial description: royal inscriptions record inter-regional distances in *bēru*, and *kudurru* boundary stones deposit territorial extents as legally authoritative records (Robson, 2008; Powell, 1987). Meridian transit observation and eclipse timing are sufficient to derive inter-site angular separations at 1° resolution (Hunger & Steele, 2019; Neugebauer, 1975).

Why 0.1 *bēru*? The 3° (0.1-*bēru*) grid spacing yields 120 equally spaced harmonic nodes covering the full 360° longitudinal domain. Coarser spacings are uninformative and finer ones produce no signal (§5.1). The choice of a base-10 fraction of the attested *bēru* is motivated by uniform divisibility, not by evidence for a discrete 3° unit in the cuneiform record (Ruggles, 1999; Silva, 2020).

Metrological coincidence with tier boundaries. Thureau-Dangin (Thureau-Dangin, 1921, p. 133) derives the linear distance-*bēru* from 180 *aslu* of 59.4 m, giving ≈10,692 m; at 111 km/° this is ≈0.00321 angular-*bēru*. Two parasangs equal one distance-*bēru* (Powell, 1987, p. 467) (≈0.00160 angular-*bēru*), and one barid equals two distance-*bērus* (Burton, 1893, vol. 2, p. 63, n. 2) (≈0.00642 angular-*bēru*). These values are numerically equal (to rounding) to the 0.25 $\hat{\sigma}$ /0.50 $\hat{\sigma}$ /1.00 $\hat{\sigma}$ tier boundaries derived from the von Mises fit (§3.3), and are presented as a historical coincidence, not as the primary derivation.

B Keyword Audit Summary

Full site-by-site audit files are indexed in Supplementary S1 (supplementary/Supplementary_S1.md; Tenenbaum 2026); regenerate all with `bash tools/update_all_audits.sh`.

B.1 Dome/Spherical Monument Filter (Tests 2, 2b, 2x, 2bx)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site’s name, XML description, and extended OUV text. All 90 matching sites are included in Test 2 without context gating.

For Test 2x, ambiguous keywords (*dome*, *domed*, *domes*, *spherical*) are subject to a two-gate context check: Gate 1 (28 negative patterns) rejects non-architectural uses; Gate 2 (86 positive patterns) requires architectural confirmation. Unambiguous keywords (*stupa/stupas*, *tholos*) are accepted without validation. The seven rejected

sites are: Hiroshima Peace Memorial, Richtersveld, Gyeongju, Viñales Valley, Rock Islands Southern Lagoon, Uluru-Kata Tjuta, and Diquís Stone Spheres.

Test 2b adds earthen mound-stage keywords (*tumulus*, *tumuli*, *barrow*, *barrows*, *kofun* unambiguous; *mound* context-validated) to trace the hemispherical tradition from burial mound through stupa to architectural dome. Test 2bx applies the same architectural context gates to the extended population. Statistics for all four variants are reported in §4.3 and §4.4.

B.2 Founding-Origin Classifier (Test 5, Post-Hoc)

Results. The classifier finds 33.0% of A+ sites carry a founding keyword versus 28.7% in the full corpus (Fisher OR = 1.25, $p = 0.209$ ns). The A++ sub-population (44 sites) shows a stronger signal: 47.7% carry a founding keyword (OR = 2.37, $p = 0.005$ **), consistent with the most precisely aligned sites being disproportionately historically foundational monuments.

The classifier assigns sites to five founding-related categories; full keyword definitions and the site-by-site listing are in the supplementary archive (Tenenbaum, 2026).

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Data Availability Statement

All data and code are openly available; a permanent snapshot (v1.5.0) is deposited at Zenodo (Tenenbaum 2026; <https://doi.org/10.5281/zenodo.19868418>); UNESCO XML: UNESCO World Heritage Centre 2024. Scripts, audit files, and maps are indexed in Supplementary S1 (supplementary/Supplementary_S1.md), with each file’s role, macros produced, and paper section noted. Full reproduction: `reproduce_all_macros.sh`, `update_all_audits.sh`, `generate_figures.py`.

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Conflict of Interest

The author declares no conflict of interest.

Software and AI Disclosure

All analyses were performed in Python 3 with NumPy, SciPy, and pandas. Claude (Anthropic) assisted with manuscript drafting and prose editing. GitHub Copilot (Microsoft) assisted with code autocompletion. All scientific conclusions, hypothesis design, keyword-list decisions, and analytical choices are solely the author’s.